Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Bateman, Ted A. Ph.D.		
Project Title:	Examination of Anti-Resorptive and Anabolic Treatments/Stimuli on Unloading Induced Osteoporosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27599	**Congressional District:**	4
Comments:	Previous affiliation was Clemson University; PI moved to UNC in fall 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed end date to 9/30/2008 (from 7/31/2008) per K. Major/NSBRI (6/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Simske, Steven (University of Colorado) Norrdin, Robert (Colorado State University)		
Grant/Contract No.:	NCC 9-58-BL00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Studies examining the calcium balance during extended spaceflight on the Russian Space Station Mir and the International Space Station suggest that 1) both bone resorption and formation are altered to adversely affect bone density, and 2) the time required to return bone mass to pre-flight levels may be considerably longer than the period of microgravity exposure. Countermeasures to prevent the loss of bone mass, and therapies to more quickly replace bone post-flight, are desirable to not only prevent fractures during long-duration missions, but to also prevent microgravity-associated bone loss from exacerbating naturally occurring osteoporosis. Given the severity of bone loss, the long-term health concerns, and the numerous biomedical problems associated with spaceflight, every effort should be made to develop countermeasures that require minimal intervention.
In the first year of this grant we developed a treatment regimen that would prevent in-flight bone loss and accelerate the recovery of bone health post-flight. Both bone mass and turnover rates are important for evaluating successful recovery. The significance of these factors is illustrated by our demonstrating that low doses of anti-resorptive agents can prevent in-flight loss of bone mass and allow normal bone formation rates. Through a series of iterative studies in mice which were administered either osteoprotegerin (OPG) or zoledronate, doses of 500 µg/kg for OPG and 45 µg/kg, respectively, were determined to be the minimum, efficacious comparable doses in that they increased bone mass, while not causing the inhibition of bone formation that generally accompanies antiresorptive therapies.

These doses of OPG and zoledronate were then tested for their ability to prevent disuse osteoporosis with the hindlimb suspension model in mice. Both antiresorptive therapies, at these lower doses, maintained bone mass, strength and structural properties at the level of fully loaded, placebo treated control mice. This study demonstrates the potential benefits of minimum dosing. A manuscript reporting both aims of this “proof of concept” approach to developing and testing minimum, efficacious therapies is in revision for the.

The most significant finding reported in this paper is the potential synergistic inhibition of bone formation caused by the combination of antiresorptive therapies and disuse. Though both doses of OPG and zoledronate caused no inhibition of bone formation in normally loaded control mice, the antiresorptive therapies caused an inhibition of bone formation of 42-44% in the hindlimb unloaded mice in addition to the 45% inhibition of bone formation caused by hindlimb suspension with no drug treatment. Though this observation is in mice, the potential synergistic inhibition of bone formation should be examined in astronauts as bone loss therapies are introduced.

The remaining stages of this project are examining the sequential administration of bisphosphonates and anabolic PTH therapy. This is being done in mice, using the hindlimb suspension model, to determine if anabolic treatment after disuse can reverse the inhibition of bone formation that both unloading and antiresorptive therapy cause. We have also modified this aim to include a clinical examination of sequential antiresorptive and anabolic therapies in women receiving treatment for osteoporosis. Preliminary data collected in 105 patients suggest that prior bisphosphonate treatment inhibits the anabolic action of PTH. Our retrospective chart review has been expanded to include up to 600 patients, allowing a more definitive determination of the efficacy of sequential therapy and potentially discriminate differences between bisphosphonates.

In addition to altered gravity, astronauts on exploratory missions will be exposed to space radiation. We have grown the scope of our lab to include examining the effects of space relevant types of radiation, and have identified bone loss at doses relevant to these missions. This work is being performed by combining resources from an NSBRI (one year Synergy grant completed in 2006), Procter and Gamble Pharmaceuticals, South Carolina Space Grant Consortium, and NASA/SC EPSCoR grants and a NASA GSRP Fellowship through Kennedy Space Center. This is a significant additional focus, but NASA’s Exploration Roadmap has also changed with an increased emphasis on lunar exploration. The inclusion of space radiation-induced bone loss to our research fits this change in the Roadmap.

We have reported (Journal of Applied Physiology, 2006) a significant loss of bone mass at a dose, 2 Gray (Gy), lower than previously reported and approaching what astronauts will experience on the moon and Mars. Declines in trabecular volume fraction were observed for all radiation species four months after exposure: gamma, (-29%), proton, (-35%), carbon (-39%), and iron (-34%). These data suggest that space radiation has the potential to exacerbate the bone loss caused by microgravity.

A one year pilot grant from NSBRI funded an examination of the dose threshold for bone loss caused by proton radiation (modeling a solar flair), examining doses of 2, 1 and 0.5 Gy four months after exposure. This project concluded that the threshold for significant bone loss is between 0.5 and 1 Gy, a dose astronauts may possibly receive during a lunar mission and are likely to receive on a mission to Mars. These data have been submitted for publication. Three other papers have been submitted reporting aspects of this research, and suggest that bone loss from moderate doses of radiation may be rapid.

In addition to a new award from NSBRI to examine space radiation-induced bone loss over the next four years, we have received a two-year grant from NIH to develop an animal model for radiation therapy induced bone loss. This work represents a “NASA Spin-off” that may result in the prevention of fractures in cancer patients.

Rationale for HRP Directed Research:

The human skeletal system consists of a dynamic balance between solid, calcified bone and free Ca2+ in the blood. Two processes, resorption and formation, consistently turnover bone and contribute to “bone remodeling”. An imbalance in this system, such as in post-menopausal women or during spaceflight, can result in a shift favoring bone resorption. The result of this imbalance is low bone density, bone mass and an increase risk of fracture.
Research in the area of spaceflight induced bone loss has implications for the causes of osteoporosis on earth, and in particular the development of effective therapies to prevent and reverse bone loss. Indeed, studies on the International Space Station reveal that bone loss occurs at a rate much faster than of post-menopausal woman. With over 1.5 million fractures occurring per year due to the proliferation of an aging population and the expansion of space exploration to unprecedented new lengths, the need for the development of a variety of advanced, effective therapies has never been greater.

Both astronauts as a population and spaceflight as a cause are significantly different than the population and pathology current osteoporosis therapies were designed to treat. It should not be presumed that doses and therapies intended for older women are appropriate and efficacious for astronauts on exploratory missions. Optimizing treatment regimens for populations, like astronauts, may eventually be expanded to modifying regimens to individuals, depending on specific physiological deficits and response.

Research Impact/Earth Benefits:	A better understanding of radiation-induced bone loss is a significant clinical topic that is largely unexamined. A recent retrospective analysis of more than 6,000 postmenopausal women receiving radiotherapy for cervical, rectal, and anal cancers established a significant correlation between radiation therapy and increased fracture risk. Risk of hip fracture among these patients was elevated five years after therapy (relative risks of 1.66, 1.65 and 3.16, respectively). Skeletal complications from radiation therapy have been described for breast, brain, and pelvic cancers and leukemia (bone-marrow transplants). Improved survivorship rates of cancer patients receiving radiotherapy increase the importance of understanding the mechanisms and long-term effects of radiation-induced bone loss. Nether cellular nor molecular causes for this skeletal deficit caused by radiation therapy are appropriately understood. Radiation therapy is administered as whole-body (preceding bone marrow transplant) or local, targeting the tumor. Protocols vary considerably, but generally whole-body irradiation totals 10-15 Gy and is administered in daily fractions of 1-2 Gy, whereas local therapy totals 40-70 Gy delivered in approximately 2 Gy fractions. Understanding the role of radiation therapy in fracture risk is complicated by the fact that this bone loss is often accompanied by several contributors to low bone density, including hormone changes, steroid administration, reduced activity levels, altered diet, and chemotherapy. Identifying the independent effects of radiation on the skeletal system requires an animal model. To date, one does not exist. Further development of this model is necessary, including the characterization of the time course of the loss, the potential sparing effect of fractionation and low-dose rate, and identification of dose thresholds, so that molecular mechanisms and therapies can be examined. In addition to a new award from NSBRI to examine space radiation-induced bone loss over the next four years, we have received a two-year grant from NIH to develop an animal model for radiation therapy induced bone loss. This work represents a “NASA Spinoff” that may result in the prevention of fractures in cancer patients.
Task Progress:	Aim 1 (completed): In the first year of this grant we developed a treatment regimen that would prevent in-flight bone loss and accelerate the recovery of bone health post-flight. Through a series of iterative studies in mice administered either osteoprotegerin (OPG) or zoledronate, doses of 500 µg/kg for OPG and 45 µg/kg, respectively, were determined to be the minimum, efficacious comparable doses: they increased bone mass, while not causing the inhibition of bone formation that generally accompanies antiresorptive therapies. Aim 2 (completed): These doses of OPG and zoledronate were then tested for their ability to prevent disuse osteoporosis with the hindlimb suspension model in mice. Both antiresorptive therapies, at these lower doses, maintained bone mass, strength and structural properties at the level of fully loaded, placebo treated control mice. This study demonstrates the potential efficacy of minimum dosing. The most significant finding from hindlimb suspension study is the potential synergistic inhibition of bone formation caused by the combination of antiresorptive therapies and disuse. Though both doses of OPG and zoledronate caused no inhibition of bone formation in normally loaded control mice, the antiresorptive therapies caused an inhibition of bone formation of 42-44% in the hindlimb unloaded mice in addition to the 45% inhibition of bone formation caused by hindlimb suspension with no drug treatment. Aim 3: The remaining stages of this project are examining the sequential administration of bisphosphonates and anabolic PTH therapy. This is being done in mice, using the hindlimb suspension model, to determine if anabolic treatment after disuse can reverse the inhibition of bone formation that both unloading and antiresorptive therapy cause. We have also expanded this aim to include a clinical examination of sequential antiresorptive and anabolic therapies in women receiving treatment for osteoporosis. Preliminary data collected in 105 patients suggest that prior bisphosphonate treatment inhibits the anabolic action of PTH. Our retrospective chart review is continuing and now includes 600 patients. Aim 4: In addition to altered gravity, astronauts on exploratory missions will be exposed to space radiation. We have reported a significant loss of bone mass at a dose, 2 Gray (Gy), lower than previously reported. Declines in trabecular volume fraction were observed for all radiation species four months after exposure: gamma, (-29%), proton, (-35%), carbon (-39%), and iron (-34%). These data suggest that space radiation has the potential to exacerbate the bone loss caused by microgravity. We have performed examination of the dose threshold for bone loss caused by proton radiation (modeling a solar flare), examining doses of 2, 1 and 0.5 Gy four months after exposure. This project concluded that the threshold for significant bone loss is between 0.5 and 1 Gy, a dose astronauts may receive on exploratory spaceflight missions.
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